Lecture 17

2/27/26
JRA

LSCM - remove out-of-focus light w/ pinhole in intermediate image plane

Illuminate point-by-point w/ focused laser
⇒ slow data collection (~1 sec to collect one optical section (i.e., 2D image at one focal plane)

Can we speed things up? Yes, w/ multi-spot illumination!

Today: Finish up well-established methods to remove blur in thick bio samples
3D samples

Bio samples all 3D
Take sections
Rebuild using computer

Problem: images tend to be blurry

Looked at Lily image

Common methods to remove blur

If choose to illuminate sample point by point
Put pinhole in front of detector in at image plane
Light from plane in sample that's in focus will get thru PH

Light from above & below focus will be blocked
because blurry at PH
Preferentially block OOF light

Scanning is slow ~1 sec/image
or optical section
(2D image) jargon at one focal plane

Soln is SDCM

Spinning Disk Confocal Microscopy

This technique uses multi-beam spot illumination
to speed things up

This comes at a cost

Can do real-time imaging 30 frames/sec or faster
Limit is getting sufficient signal strength

How does this work?

See PP image

Disk w/ many pinholes
Expand laser beam to hit about 1000 PHs at a time

Disk is in an image plane
Disk is imaged onto sample so you have ~1000
illuminated points at the same time

Essence of SDCM

has lots of holes
(1) Illum ~1000 holes in a disk simultaneously
Disk is simultaneously in focus w/ sample.
(like the field diaphragm

⇒ lots of illumination spots are focused
onto sample (by objective)
instead of one as in LSCM

This makes technique <u>much</u> faster

(2) We still need to remove blur (OOF Fluorescence)

Because pinhole wheel is in an image plane,
the pinhole wheel blocks OOF light. Movement
of PW not an issue because abs/em & subsequent flt so fast

Sample (spots) Fluoresce and objective collects
Fluorescence.

Same pinhole used to make spot collects light from
spot. Same principle as LSCM. It passes thru PH

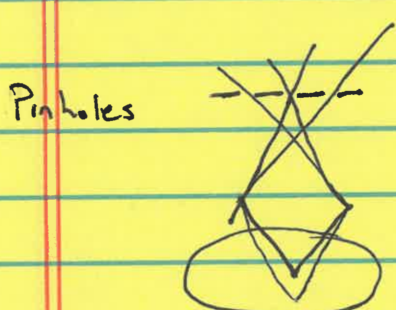
See PP figure w/ thin & thick samples

3. What's the rub?

For thicker samples, you get pinhole crosstalk that reduces deblurring effect by reducing rejection of OOF light, so images can be less effectively deblurred.

Look at PP figure. OOF Light sneaks thru nearby PHs.

One pinhole illuminates
Many pinholes collect



No problem of LSCM because only 1 PH

Get OOF light sneaking thru 3 PHs. Crosstalk. Increased blur.

This is the method of choice for Fast dynamic processes.

This depends on how fat sample is

~~VIDEO~~
ANIMATION

Drawbacks.

Multiple PHs ē cameras

Because illuminating lots of spots
→ use camera

Disk PHs in spiral. 5000 rpm spin disk

Illum entire sample at 33 msec std frame rate

EX] Drosophila GFP-H2B
mCherry-tubulin

S. cerevisiae RFP Z stack of mito

Looked at sample SD

LSCM

1. Single point illum
2. Single filtering PH to remove blur in int image plane
3. Laser illumination
4. Raster scan point by point
5. Excellent OOF rejection
6. Slow
7. PMT for detection NOT efficient photon detectors Fast ~25%
8. Image is assembled by computer

Never see an image thru eyepieces

SDCM

1. Multibeam Pinhole illumination using a disk
2. OOF light rejected by same pinhole as used for illum (BUT cross talk) PH excites & rejects
3 Lasers 4. Disk spins Disk spins to scan sample
5. Good out-of-focus rejection for thin samples but PH cross talk bad if sample is thick
6. Fast
7. Camera for detector since illuminating more sensitive 90% eff
8. Image obtained

Underlying theme – use PH to reject OOF light

Decon

Computer-based Deblurring Method

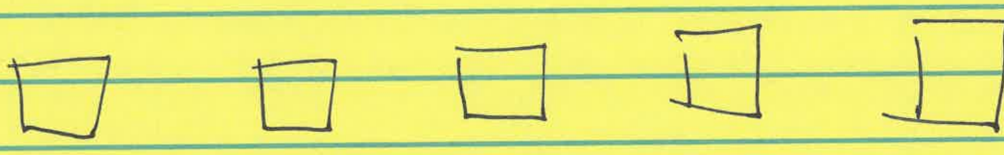
There is another technique for deblurring fluorescence imaging thats entirely computer based. Do this w/o a PH.

Use std WF fluorescence μscope.

Download Fiji

Look at PSF

Take tiny fluorescent beads.
Image in different planes



Looked at stack of 40 images taken at different z heights

Every image is a superposition of pts

Can use computer techniques to deblur image

If computer knows the PSF (which can be measured) the computer can computationally deblur image

PSF = 3D image of a pt source of light

No HW Computer uses PSF

HW Look at PSF in ImageJ. See light OOF

Total F is always the same

IF Bright, compact
OOF Dim, spread out

This is what's expected from ray tracing